A Structural Derivation of the Consciousness Dynamical System (CDS)

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Abstract

This paper proposes a formal mathematical framework for consciousness grounded in a set of foundational postulates. Within this framework, consciousness is characterized as a formalizable dynamical system (hereafter referred to as the Consciousness Dynamical System, CDS). Under a unified variable scheme, the system's core equation—termed the consciousness dynamical equation—is derived:

$$C = \frac{fd}{a + l}$$

Here, C denotes coherence intensity. The variables f, d, a, and l represent the potential structural field, dynamical drive, information gradient, and information manifold, respectively. The term fd characterizes the generative contribution of the system, while $a + l$ expresses the constraint and normalization conditions governing this generation. When the steady–state closed–loop condition $a + l \neq 0$ is satisfied, the equation corresponds to the consciousness actualized state equation defined in this framework.

This study addresses the following central question: Can consciousness be axiomatically characterized under minimal structural prerequisites and, from there, yield formal dynamical laws?

The main contributions of the paper are as follows:

(1) Research Objective: To construct a formalizable and derivable mathematical representation of consciousness;

(2) Methodology: Postulate–based modeling, variable decomposition, and formal system derivation;

(3) Result: Derivation of the consciousness dynamical equation $C = fd/(a + l)$ under steady–state constraint conditions;

(4) Significance: The framework provides a formally mappable variable structure applicable to life–information systems, cognitive dynamics, and artificial system design.

This study does not aim to provide an empirical definition of consciousness. Instead, it offers an axiomatic theoretical pathway characterized by mathematical self–consistency, structural coherence, and potential testability.

Note: For continuity of exposition, equations are displayed in vertical form in derivation sections and presented in inline form within narrative text.

1. Introduction

1.1 Structural Difficulties in Existing Consciousness Research

Current research on consciousness can broadly be divided into two categories.

1.1.1 Philosophical Traditions

Philosophical approaches to consciousness primarily rely on conceptual analysis and phenomenological reflection. While these methods are effective in posing questions and clarifying concepts, they typically lack formalized expression, reproducible chains of derivation, and mathematically formalizable, computable, and verifiable representations.

1.1.2 Empirical Sciences (Neuroscience / Cognitive Science / Artificial Intelligence)

Empirical research commonly approaches consciousness through neural structures, behavioral manifestations, or computational models, and has achieved substantial progress in correlational analysis and functional explanation. However, at the level of underlying mechanisms, there remains a lack of formalized frameworks. Existing theories are often constructed through fragmented modeling approaches, making it difficult to form a self–consistent formal system capable of systematic derivation. Moreover, there is still a lack of a formal bridge capable of aligning structural representations with first–person subjective experience.

1.2 The Necessity of Axiomatization and Formalization

The importance of Euclidean geometry does not lie merely in its specific conclusions, but in its methodological structure: beginning from postulates and, through formal systems and deductive reasoning, achieving traceable consistency and verifiability.

Accordingly, this paper introduces an axiomatic and formal strategy:
(1) Selecting propositions that simultaneously satisfy self–evidentiality, non–refutability, universality, and necessity as foundational postulates;
(2) Translating the structure of consciousness into a formal framework of variables–relations–constraints;

(3) Adopting formal derivation rather than empirical curve–fitting.

Within this framework, the present study derives the actualized state equation of consciousness:

$$C = \frac{fd}{a + l}\ , \text{ s.t. } a + l \neq 0$$

1.3 Research Positioning and Contributions

This study does not aim to propose a new empirical definition of consciousness. Instead, it seeks to provide a formalizable structural framework: translating subjective experience from philosophical–semantic description into a computable structural object, and establishing a formal variable–mapping pathway for subsequent model derivation, explanation, and potential validation.

2. Establishment of Postulates

2.1 First Postulate: The Principle of Postulate Selection

Within the tradition of axiomatic science, the reliability of a theoretical system does not depend on empirical accumulation, but on whether a systematically derivable formal structure can be generated from a minimal set of premises. Euclidean geometry, the ZFC axiomatic system, and the Peano axioms are all typical examples of this approach.

However, the field of consciousness studies still lacks a formal mechanism for selecting the minimal necessary premises. To address this gap, this paper proposes a meta–level foundational postulate—the Principle of Postulate Selection—which specifies the formal eligibility criteria by which candidate propositions may enter the axiomatic system of consciousness.

2.1.1 The Four Conditions of the Postulate Selection Principle

Let P be a candidate proposition. Define the postulate–qualification operator:

$$\mathrm{Qua}(P) := \mathrm{Self}(P) \ \wedge\ \mathrm{NonRef}(P) \ \wedge\ \mathrm{Uni}(P) \ \wedge\ \mathrm{Nec}(P)$$

A proposition P is eligible for inclusion in the axiomatic system of consciousness if and only if

$$\mathrm{Qua}(P) \;=\; \mathrm{True}$$

This criterion consists of four jointly necessary and sufficient conditions: self–evidentiality, non–refutability, universality, and necessity.

Here, Qua (Qualification) denotes the postulate–qualification operator, used to determine whether a proposition P satisfies the postulate–selection criteria. Self, NonRef, Uni, and Nec correspond to the four selection conditions, whose conceptual naming rules are provided in Appendix A.2.

The symbol := denotes defined as, and ∧ denotes logical conjunction (and). The syntactic rules for the mathematical symbols are provided in Appendix A.4.

(1) Self–Evidentiality

A postulate must be directly evident at the level of logic, subjective experience, or structural necessity, without relying on an external inferential system. The criterion is:

$$\mathrm{Self}(P) = \mathrm{True} \Leftrightarrow (\neg P \vdash \perp)$$

That is, the negation of P leads to a logical contradiction; moreover, this negation process inevitably presupposes and employs the structure of P itself, thereby forming a self–validating closure.

Here, ⇔ denotes if and only if, ¬ denotes negation, ⊢ denotes derivable, and ⊥ denotes logical contradiction.

(2) Non–Refutability

A postulate must admit no constructible counterexample. The criterion is:

$$\mathrm{NonRef}(P) = \mathrm{True} \Leftrightarrow \neg\exists x \in D : \mathrm{Counterexample}(x,P)$$

That is, there exists no instance x within the counterexample search domain D such that x constitutes a counterexample to P. If a counterexample exists, then P is closer to an empirical regularity or a local factual proposition rather than a true postulate.

Here, D denotes the counterexample search domain, and x denotes a candidate instance within that domain.

The symbol ∃ denotes there exists, ∈ denotes belongs to, and : denotes such that (set–builder notation).

(3) Universality

A postulate must remain valid across different types of subjects. The criterion is:

$$\mathrm{Uni}(P) = \mathrm{True} \Leftrightarrow \forall s \in S : P(s)$$

where:

S **:=** { s | s is a human–level subject, an animal subject, an AI system, or another possible life form }

That is, the proposition P holds for every subject s within the subject–domain S. If P applies only to specific subjects, it is closer to a local constraint statement rather than a postulate.

Here, S denotes the subject–domain under discussion, and s denotes an individual subject (or system instance). This symbol convention will be used throughout the remainder of the paper.

The symbol ∀ denotes for all.

(4) Necessity

A postulate must constitute a minimal necessary condition for the well–formedness of the system. The criterion is:

$$\mathrm{Nec}(P) = \mathrm{True} \Leftrightarrow \neg\mathrm{WellFormed}(\mathrm{System} \setminus \{P\})$$

That is, there exists no system excluding P that still satisfies the condition of well–formedness.

Here, System denotes the axiomatic system proposed in this paper, and WellFormed(·) denotes the predicate that determines the formal well–formedness of a system (i.e., the system remains syntactically, structurally, and inferentially coherent and well–defined).

2.1.2 Significance of the Postulate Selection Principle

By introducing an external meta–level screening criterion, the Principle of Postulate Selection provides a structured entry condition for subsequent formal derivation. This reduces the risks of conceptual circularity and rhetorical argumentation, ensuring that propositions entering the system possess formal self–sufficiency.

Note: This principle does not provide specific theoretical content; rather, it establishes the boundary of formal legitimacy for the generative process of the theoretical structure.

2.2 Second Postulate: The Fivefold Law of Reality

Before constructing an axiomatizable system of consciousness, a prior foundation must be established: whether the reality upon which consciousness depends (Reality, denoted R) is itself valid—that is, whether "truth" exists at all. Without a foundational reality that is non–deniable, non–extrapolative, and independent of internal system derivation, any discussion of consciousness would inevitably fall into circular grounding, system self–reference, or Gödelian closure. The system would be unable to provide an absolute foundational starting point from within itself.

Furthermore, if one asserts that "truth does not exist," this assertion itself must still function as a truth–evaluable proposition; otherwise, it loses its status as a proposition altogether. Hence, the existence of truth is not an empirical claim, but a structural precondition.

To establish an external anchor and avoid foundational regress, this paper draws inspiration from Parmenides' conception of Being as fundamentally self–grounding, and abstracts this insight into a formal structural formulation. We thus propose the Second Postulate—The Fivefold Law of Reality—Any structure that may be called a reality–level structure must satisfy topological closure and logical independence, and can be formally characterized by the following five jointly necessary and sufficient constraints.

2.2.1 The Five Conditions of the Fivefold Law of Reality

In this paper, Reality (R) is defined as a structure that simultaneously satisfies five necessary and sufficient conditions: uniqueness, infinitude, duration, constancy, and non–otherness.

(1) Uniqueness

If R were definable by a higher–order entity, then that higher–order entity would constitute a more fundamental reality. Therefore, reality cannot be externally determined:

$$\mathrm{Uni}(R)\text{: } \neg\exists X \in Ext,\ \exists f,\ R = f(X)$$

That is, there exists no external condition X and mapping relation f such that R is completely determined from outside.

Here, Ext denotes the domain of external structures, X an external candidate determinant, and f a structural generative or mapping relation.

(2) Infinitude

If R had a boundary, then beyond that boundary there would exist a larger domain capable of describing it, thereby negating its status as ultimate reality. Thus, reality must be a total

structure with no exterior:

$$\mathrm{Inf}(R) := U \setminus R = \varnothing \ (\Leftrightarrow R = U)$$

That is, there exists no object belonging to the universal domain U that lies outside R.

Here, U denotes the universal domain of describable entities.

The symbol \ denotes set difference, and $\varnothing$ denotes the empty set.

(3) Duration

If R changed over time, there would necessarily be a higher–order standpoint distinguishing an "old reality" from a "new reality," and thus reality would no longer be fundamental. Therefore, reality must be temporally invariant:

$$\mathrm{Dur}(R)\text{: } \forall t_1,\ t_2,\ R(t_1) = R(t_2)(\Leftrightarrow dR/dt = 0)$$

That is, there exists no time parameter t under which the value of R varies.

Here, t_1, $t_2$denote arbitrary time parameters, d denotes an infinitesimal change (used to express rate of variation), and $R(t)$ denotes a time–parameterized representation of R.

(4) Constancy

If R varied with observer, perspective, or measurement conditions, it would be merely an observational product rather than a structure of reality itself. Thus, reality must preserve representational invariance:

$$\mathrm{Con}(R)\text{: } \forall o,v,\ \mathrm{Rep}(o,R) = \mathrm{Rep}(v,R)$$

That is, no change in observational conditions may invalidate the invariance of R's representation.

Here, $\mathrm{Rep}(\cdot)$ denotes the representational mapping of structure R under given observational conditions; o denotes variation in observing subjects, and v denotes variation in perspective, scale, or measurement conditions.

(5) Non–Otherness

If the truth of R depended on an external grantor of validity, then that external entity would be more fundamental. Therefore, reality must be structurally self–grounded:

$$\mathrm{NonOther}(R)\text{: } \neg\exists Y \neq R,\ \mathrm{Grounds}(Y,R)$$

That is, there exists no entity Y distinct from R that serves as the ultimate grounding source of R.

Here, Y denotes a candidate grounding entity distinct from R, and Grounds(Y, R) denotes that Y provides the ultimate ontological ground of R.

These five conditions are not empirical definitions, but a set of necessary constraints determining which structures qualify as reality–level structures for subsequent formal derivation.

2.2.2 Postulate Qualification: Satisfaction of the Four Selection Conditions

(1) Self–Evidentiality

To deny the existence of reality is to assert "reality does not exist." Yet this assertion must itself function as a truth–evaluable proposition; otherwise, it loses propositional status. Thus, the denial of reality presupposes the existence of a truth–structure, forming a self–validating closure. Therefore, "reality necessarily exists," satisfying the condition of self–evidentiality.

(2) Non–Refutability

To refute the Fivefold Law of Reality, one would need to construct an object that violates at least one of the five conditions while still being called "reality." However:

- Any structure describable or validated from outside violates uniqueness or non–otherness;
- Any structure with a boundary violates infinitude;
- Any structure dependent on temporal evolution violates duration;
- Any structure varying with observation violates constancy.

Hence, no counterexample can satisfy the status of reality while violating the Fivefold Law of Reality. The condition of non–refutability is satisfied.

(3) Universality

The Fivefold Law of Reality does not depend on specific disciplines or cultural frameworks. Whether in ontology, set theory, topology, systems theory, or model theory, once an object is elevated to a "reality–level concept," its structure inevitably falls under these five constraints. Thus, universality is satisfied.

(4) Necessity

Without the Fivefold Law of Reality, one cannot distinguish between reality, observation, and local fact, nor can one establish a non–circular axiomatic system or an external anchor. The system would collapse into internal self–reference and be unable to enter rigorous formal derivation. Therefore, the Fivefold Law of Reality is a necessary condition for the viability of the axiomatic system of consciousness.

2.2.3 Significance of the Fivefold Law of Reality

The Fivefold Law of Reality does not directly answer what reality is, but rather what kind of structure is eligible to be called reality. Through five formal constraints, *R* is restricted to a structural foundation that:

- does not depend on external validation;
- is not enveloped by a higher–order system;
- does not fluctuate with time or observation;
- has no external boundary;
- cannot be further extrapolated outward.

Thus, the Fivefold Law of Reality provides a non–circular, non–regressive, and externally stable foundational point for the axiomatic system of consciousness, preventing the system from collapsing into internal paradox and supplying the outermost coordinate origin for the subsequent formal derivation of consciousness models and equations.

2.3 Third Postulate: Cogito, ergo sum

In the philosophical tradition, *Cogito, ergo sum* ("I think, therefore I am") has long been regarded as one of the strongest self–evident propositions. Its force lies in the fact that any act of doubt necessarily presupposes a subject–event of "doubting" or "experiencing." Thus, this proposition is not an empirical judgment but a mode of existence that cannot be negated: it does not depend on whether the external world is real, but only on the structural fact that some experiential event is occurring.

However, when consciousness is reconstructed in a formal system, the linguistic proposition "I think, therefore I am" cannot directly enter formal derivation. Therefore, this paper translates it into a symbolizable, operable relational structure and establishes it as the Third Postulate—Cogito, ergo sum.

2.3.1 Formal Treatment of "Cogito, ergo sum"

(1) Introduction of Variables

To avoid ambiguity in terms such as "thought" "mind" or "consciousness," we minimally define the core elements as follows:

- *i*—a subjective experiential event (an irreducible awareness / experience occurring at the experiential level);
- *c*—a consciousness constraint (the structural precondition that allows experience to appear and be accessible);
- *E*—existence (the structural manifestation of experience under constraint).

Here, *i* does not refer to personality, narrative selfhood, or memory systems, but solely to the minimal unit: an experience is occurring.

(2) From "Thinking" to "Subjective Experience"

Descartes' "thinking" does not primarily refer to reasoning, language, action, or memory, but to a more fundamental fact: the subject "I" is aware that awareness is occurring. In contemporary consciousness research, this corresponds to a subjective experiential event.

Formally:

$$i = \text{subjective experiential event}$$

Importantly, *i* does not describe experiential content, but the fact that an experiential event is taking place.

(3) From "Existence" to "Consciousness Constraint"

"Cogito, ergo sum" may be decomposed into three components:

(i) "I think": a subjective experiential event is occurring;

(ii) "Therefore": this event is bound by an unavoidable logical constraint;

(iii) "I am": existence is confirmed and presented at the experiential level.

Yet the proposition does not explicitly specify the structural condition that allows experience to occur. Moreover, "thinking" is not equivalent to linguistic judgment but is an irreducible awareness–event, which minimally entails:

(i) Presence—the occurrence of experience implies some background of awareness;

(ii) Reflexive accessibility—awareness includes accessibility to itself (minimal self–reference);

(iii) Co–temporal constitution—the occurrence of experience and its being–aware occur simultaneously.

Thus,"I am" is not a self–sufficient property of an entity, but necessarily depends on the structural constraints that allow experience to appear and be accessible. Structurally, "I am" presupposes "consciousness is," that is:

$$\frac{\text{"I am"}}{\text{"consciousness is"}}$$

To enter a structural modeling context, we introduce:

$$c = \text{consciousness constraint}$$

Here, c does not represent experiential content, but the structural constraint that makes experience manifest and accessible. Therefore, c is not posed as an empirical claim, but as a necessary structural precondition.
Note: lowercase c denotes "consciousness constraint," while uppercase C later denotes "coherence intensity."

(4) Structural Form of "I am / Consciousness is"
In a structural modeling context, an experiential event cannot be self–sufficient, but must depend on a constraint structure that enables its manifestation and accessibility. Let i denote the subjective experiential event and c the consciousness constraint. Then "Cogito, ergo sum" can be formalized as the structural relation:

$$\frac{i}{c}$$

Its meaning is as follows:

- Conditionality of constitution: c is a necessary constraint for i; without constraint, no experience can appear;
- Variable–precondition distinction: i is the experiential event variable, while c is the structural precondition; the former varies, the latter provides stable constraint;
- Transformation of ontological statement: the expression describes the constraint condition for the constitution of experience, thereby translating "I am" from an ontological narrative into a structural relation.

Note: here i/c is a structural expression indicating that experiential event i is constituted under constraint c; it does not presuppose numerical division. The operator meaning is specified in Appendix B.

(5) Mathematical Definition of Existence
Accordingly, "existence" is not treated as a property of a subject, but formally defined as the structural presentation of experience coupled with constraint:

$$E = \frac{i}{c}\ , \text{s.t. } c \neq 0$$

Where

- i denotes the subjective experiential event;
- c denotes the consciousness constraint;
- E denotes "existence" under this formal definition.

2.3.2 Postulate Qualification: Satisfaction of the Four Conditions

(1) Self–Evidentiality

As long as a subjective experiential event i is occurring, there must exist some constraint c that makes it manifest and accessible, thereby confirming the structural existence:

$$E = \frac{i}{c} \text{ , s.t. } c \neq 0$$

This conclusion does not depend on any assumption about the external world, bodily existence, or narrative selfhood; as long as "experience is occurring," E holds. The condition of self–evidentiality is satisfied.

(2) Non–Refutability

To deny "Cogito, ergo sum" would require constructing a situation in which experience i occurs but constraint c does not:

$$i \wedge \neg c$$

However, without c, experience could not appear, nor be recognized as experience; thus, such a situation is structurally non–constructible. Equivalently, for any possible situation $\omega \in \Omega$:

$$\forall \omega \in \Omega: i(\omega) \Rightarrow c(\omega)$$

That is, whenever experience occurs, constraint must also be present. Whether in dreams, hallucinations, virtual realities, or simulated worlds, as long as "awareness is occurring," the structural relation holds. The condition of non–refutability is satisfied.

Here, ω denotes a possible situation (a state instance), and Ω denotes the set of all possible situations.

The symbol $\Rightarrow$ denotes logical implication, meaning that whenever the former holds, the latter necessarily follows.

(3) Universality

"Cogito, ergo sum" does not apply only to human subjects with language and narrative systems, but to any system capable of being aware of its experiential events, including infants, dreaming subjects, neurally augmented agents, and artificial systems with subjective experience. Its structural form remains:

$$E = \frac{i}{c} \text{ , s.t. } c \neq 0$$

This structure is independent of culture, language, memory, or identity. The condition of universality is satisfied.

(4) Necessity

Without introducing the existence structure

$$E = \frac{i}{c} \text{ , s.t. } c \neq 0$$

it would be impossible to explain how experience is constituted or to provide a formal entry point for subjectivity. Furthermore, subsequent derivations of steady–state systems, cognition–action coupling, and drive–constraint structures would lack a common structural reference. Thus, this postulate constitutes the minimal necessary precondition for system closure. The condition of necessity is satisfied.

2.3.3 Significance of "Cogito, ergo sum" ⇒ *E = i/c*

The function of the Third Postulate is not to answer what consciousness is, but to translate it from philosophical language into a formal, derivable structural expression, allowing it to enter a mathematical system and correspond with subsequent steady–state equations and dynamical equations of consciousness.

Thus, "Cogito, ergo sum" is no longer merely a self–evident semantic statement, but is structurally reformulated as:

$$\text{Existence} = \frac{\text{experience}}{\text{constraint}} \Rightarrow E = \frac{i}{c} \text{ , s.t. } c \neq 0$$

Its core significance lies in translating existence from an ontological property into a formally expressible structural relation, thereby providing the formal foundation for subsequent steady–state constraint structures and the consciousness dynamical equation.

2.4 Fourth Postulate: Knowing–Acting Coherence and Steady–State Stability

This postulate constitutes a structural translation of the classical Confucian cultivation principles of Unity of Knowing and Acting and Inner Stability. Specifically, under

closed–loop steady–state constraints, it is formulated as the condition under which a knowing–acting system can sustain coherent self–consistency without internal contradiction.

Western philosophy establishes the ontological status of consciousness through Cogito, ergo sum, yet it lacks a unified formal framework describing the conditions under which consciousness is generated and maintained at the experiential and dynamical levels. In contrast, Eastern traditions (e.g., Confucius and Wang Yangming) offer a structurally oriented perspective: consciousness is not merely passively given, but is confirmed and sustained through action and its feedback loop.

Accordingly, we propose the Fourth Postulate—Knowing–Acting Coherence and Steady–State Stability—as the consistency steady–state condition of a consciousness system. Under this postulate, deviations between cognitive structure and action strategy manifest as steady–state deviations. Constraining such deviations within a bounded range corresponds to enhanced coherence and sustained steady–state maintenance.

At the experiential level, what is felt as unity of knowing and acting together with inner stability corresponds to steady–state stability under closed–loop constraints.

2.4.1 Formal Treatment of "Unity of Knowing and Acting, and Inner Stability"

(1) Introduction of Variables

To avoid semantic ambiguity, we adopt a minimal set of variables:

- k—cognitive structure (structural constraint of knowing);
- π—action strategy (generative strategy of acting);
- a—steady–state deviation (attunement–related internal coherence deviation);
- l—liminal trace (non–zero residual ensuring continuity and serving as a regularization term);
- C—coherence intensity (degree of systemic self–consistency).

(2) Deviation in Knowing–Acting Coupling

When changes in cognition and action are inconsistent, the steady–state deviation is defined as:

$$a = |\Delta k - \Delta\pi|$$

Thus, a represents the systemic deviation from coherence: as a increases, coherence decreases and instability tendencies arise; as $a \to 0$, the system approaches high coherence and the steady–state limit.

Here, Δk and $\Delta\pi$ denote numerically mapped variations within a unified state space.

The symbol Δ denotes increment or change, and $\to$ denotes tends toward.

(3) Coherence Intensity and Steady–State Constraint

When cognitive structure and action strategy form a coupled feedback loop, coherence intensity C is introduced as:

$$C = \frac{k\pi}{a}$$

The numerator $k\pi$ represents the expansion term driven by knowing–acting coupling; the denominator a represents the constraint imposed by deviation on coherence.

Thus:

- larger $a \rightarrow$ weaker coherence;
- smaller $a \rightarrow$ the system approaches self–consistency.

(4) Liminal Trace Term and Well–Definedness

When $a \rightarrow 0$, using a as a denominator leads to singularities and renders the model ill–posed. To ensure continuity and computability, a liminal trace term l is introduced as a regularizing quantity, keeping the denominator non–zero:

$$a + l \neq 0$$

Here, l captures the irreducible structural traces and path–dependent residues accumulated during system operation. It ensures that even at the steady–state limit, the model remains well–defined.

(5) Steady–State Equation

The resulting steady–state equation is therefore:

$$C = \frac{k\pi}{a + l}, \text{ s.t. } a + l \neq 0$$

Its structural meaning is:

- $k\pi$—knowing–acting coupling drive term;
- $a + l$—aggregated steady–state constraint from deviation and trace;
- C—coherence intensity of knowing–acting unity (degree of systemic self–consistency).

Therefore, when knowing and acting are more aligned ($a + l$ decreases), C increases; when knowing and acting diverge ($a + l$ increases), C decreases and manifests as steady–state degradation (reduced stability). This structural relation constitutes the formal expression of "unity of knowing and acting, and the condition of inner stability."

2.4.2 Postulate Qualification: Satisfaction of the Four Conditions

(1) Self–Evidentiality

For systems exhibiting cognitive–action causal coupling, when cognitive structure k and action strategy π are inconsistent, the deviation term a increases and weakens coherence intensity C. This phenomenon can be directly confirmed through subjective experience and is consistent with frameworks in cognitive psychology, neuroscience, and control theory. The condition of self–evidentiality is satisfied.

(2) Non–Refutability

In a system possessing steady–state constraint capability and containing both a cognitive module k and an execution module π, the two necessarily form a coupled feedback loop. If $k \neq \pi$, the system generates a cognition–execution deviation represented by a, and the steady–state constraint $a + l$ structurally intervenes.

Thus, under the structural assumptions of this model, a state in which "knowing–acting conflict exists yet the system remains in perfect steady–state" is structurally unconstructible. The condition of non–refutability is satisfied.

(3) Universality

This postulate does not depend on a particular cultural context, but describes the general feedback structure of steady–state constrained systems: cognitive structure k, action strategy π, and the steady–state constraint term $a + l$ form a feedback loop:

$$C \Rightarrow k \Rightarrow \pi \Rightarrow a + l \Rightarrow C$$

Its steady–state objective may be uniformly expressed as the minimization tendency of knowing–acting deviation:

$$\min|\Delta k - \Delta\pi|$$

Thus, "Knowing–Acting Coherence and Steady–State Stability" characterizes a universal closed–loop structure of steady–state constrained systems. The condition of universality is satisfied.

(4) Necessity

If a consciousness model is to include autonomous constraint, steady–state maintenance, and self–consistent dynamic structure, it must introduce the steady–state constraint term $a + l$ to explain action correction and steady–state recovery. Without this constraint, the model degenerates into a passive input–output device and cannot characterize closed–loop constraint processes. The condition of necessity is satisfied.

2.5 Fifth Postulate: $A = 0$ as the Structural Boundary Criterion Between Living and Non–living systems

Across major theoretical traditions concerning the nature of life—including biology, physiology, cybernetics, and systems philosophy—steady–state constraint capability is commonly regarded as a core structural property of living systems. Its shared characteristic is that the system can maintain the persistence of its internal stable structure under external perturbations.

Based on this, the present work proposes the Fifth Postulate (Life–Boundary Criterion)—$A = 0$ serves as the structural boundary criterion distinguishing living systems from non–living systems.

2.5.1 Formalization and Variable Expansion

(1) Introduction of Variables

Define the system's steady–state constraint quantity as

$$A = a + l$$

where

- a—steady–state deviation (attunement–related internal coherence deviation);
- l—liminal trace (non–zero residual ensuring continuity and serving as a regularization term).

Note: In this framework, l simultaneously plays the role of a non–zero steady–state trace and a regularization term, characterizing path dependence and reachability under liminal structural conditions.

(2) Classification Criterion: Life Domain vs. Non–Life Domain

Using A as the life–boundary criterion:

- When $A = 0$, the system lacks steady–state constraint and trace–maintaining structure; it consequently undergoes a structural collapse into the non–living domain;
- When $A \neq 0$, the system possesses disturbance absorption and closed–loop steady–state maintenance capability and belongs to the life domain.

It should be emphasized that this work does not provide an ontological definition of life. Here, $A = 0$ serves purely as a structural classification criterion, distinguishing whether a system possesses a steady–state constraint structure and providing a boundary condition for subsequent formal derivations.

2.5.2 Postulate Qualification: Satisfaction of the Four Conditions

(1) Self–Evidentiality

The continued existence of a living system depends on the presence of steady–state constraint structure. This can be directly verified at multiple levels (e.g., metabolic shutdown or osmotic failure leading to the termination of life processes).

Formally,

$$A(s) = 0 \Rightarrow s \notin S_\text{life}$$

That is, when steady–state constraint fails, the subject exits the life domain.

Thus, $A \neq 0$ may be regarded as the minimal maintenance condition for membership in the life domain; when $A = 0$, this condition fails and the subject leaves the life domain. The condition of self–evidentiality is satisfied.

Here, the symbol $\notin$ denotes is not an element of (a set).

(2) Non–Refutability

To refute this postulate, one must construct a counterexample: a system that can still be classified as living despite completely lacking steady–state constraint structure.

Formally,

$$\exists\ s \in S_\text{life},\ s.\text{t}\ A(s) = 0$$

That is, there exists a subject still classified as living whose steady–state constraint has entirely failed ($A = 0$).

Within observable biological domains and existing synthetic biology or artificial life models, no evidence supports such a counterexample. Thus, in the empirical life domain, a counterexample is difficult to construct. The condition of non–refutability is satisfied.

It should be noted that within this formal framework, the criterion is adopted as a minimal necessary structural condition; stricter proofs may be provided in future formal or engineering systems.

(3) Universality

Although different theoretical traditions do not agree on a single definition of "life," there is stable consensus that steady–state constraint forms the structural basis of living systems: life necessarily involves feedback–loop structures that maintain internal stability.

Formally,

$$\forall s \in S_\text{life},\ A(s) \neq 0$$

That is, any system classified within the life domain must contain a steady–state constraint structure that maintains internal stability ($A \neq 0$).

Therefore, this postulate has consistent applicability across species, hierarchical levels, and implementation pathways. The condition of universality is satisfied.

(4) Necessity

If the steady–state constraint A is removed from the formal system, one can no longer provide a structural distinction between the life and non–life domains. This leads to:

- the loss of steady–state constraint in the knowing–acting coupling term;
- the absence of a boundary reference in the consciousness dynamical equation;
- and the failure of closure (well–formedness) in the overall dynamical model.

Formally,

$$A \neq 0 \Rightarrow \text{living system, and } A = 0 \Rightarrow \text{non–living system}$$

Thus, whether A equals zero constitutes the structural boundary criterion between living and non–living systems.

Therefore, A is a necessary condition for the establishment of the life domain.

3. Structural Derivation of the Consciousness Dynamical System (CDS)

3.1 Structural Sufficiency Proof for the Existence of Consciousness

Before formally modeling consciousness, a minimal premise must be established: consciousness is confirmably existent, and its existence possesses formalizable structural conditions. Otherwise, the formal system would lack referential grounding or an empirically confirmable anchor. The following conclusions serve as the logical starting point for subsequent model derivations.

3.1.1 Consciousness Necessarily Exists

The self–evidential nature of *cogito ergo sum* does not rely on empirical facts but on structure: any act of doubting, experiencing, or being aware already constitutes a consciousness event. The external world, perception, memory, and even the body may all be doubted; what cannot be denied is the occurrence of a presently occurring subjective experience event.

Therefore, external objects may be doubted, whereas a consciousness event—as an experience occurring in the present—is indubitable.

3.1.2 Consciousness Satisfies the "Fivefold Law of Reality"

The Second Postulate (Fivefold Law of Reality) establishes that "reality necessarily exists" as a structural necessity. Furthermore, within the domain of confirmable entities, consciousness occupies a unique position: it is not an inference or external report but the first–person present "I am," constituting an experiential anchor that is directly confirmable without external verification.

Under the criteria of the Fivefold Law of Reality, consciousness satisfies all five conditions:

(1) Uniqueness—For any subject, "my consciousness" cannot be defined by any higher–order external object; its grounding lies in the irreplaceability of first–person experience;

(2) Infinitude—Consciousness can generate and sustain experiential structures beyond the present physical coordinates; its reachable domain is not closed by the current physical reference frame (dreams, imagination, memory, future simulation, etc.);

(3) Duration—All experiences appear in the form of "present experience" and are carried by the continuity and coherence of their temporal structure;

(4) Constancy—Experiential content may vary, but the structural form "there is ongoing experience" remains invariant;

(5) Non–Otherness—Consciousness does not depend on external objects for truth attribution; its validity is grounded in intrinsic self–evidence.

Thus, within the present postulate system, consciousness satisfies all criteria of a reality–level structure, yielding the internal structural equivalence:

$$\text{Consciousness} \equiv \text{Reality–Level Structure } (R)$$

That is, within this formal framework, consciousness—as the self–evident structure of first–person experience—occupies the same structural role as R.

In this system, "reality" is defined as the total foundational structure that allows existence to appear (i.e., the basis upon which anything can be established). From this, the internal system inference follows:Where there is consciousness, there is being. Consciousness is not a part of the world, but the structural precondition for the world to arise.

It must be emphasized that this inference serves only for structural closure within the formal system and is not asserted as an empirical proposition.

Here, the symbol ≡ denotes structural identity equivalence, meaning that within the definitional domain of this formal system, the two assume the same structural role.

3.1.3 The Ternary Structure of Consciousness (Minimal Structural Unit)

If consciousness constitutes an experiential and formalizable system, it must satisfy three necessary and sufficient structural conditions:
(1) There exists a subject undergoing experience;
(2) The subject possesses the capacity for conscious activity;
(3) Conscious activity can be directed toward and capture an object.

Formally expressed as:

$$\mathcal{E}:=(S, Act, O)$$

Any event that can be constituted as a "conscious experience" $\mathcal{E}$ must structurally include the subject term S, an action term Act, and the object term O. These three form an irreducible minimal ternary coupling unit.

Where:

- $\mathcal{E}$—a consciousness experience event, the minimal structural representation of a single conscious experience;
- S—the experiential subject term, the bearer in whose presence the experience occurs;
- Act—the conscious action term, the intentional activity through which the subject directs and unfolds experience;
- O—the object term captured, the content toward which conscious activity is directed.

Subject, action, and object together constitute the minimal complete ternary structure of the consciousness system, forming the foundational unit for subsequent layered structural analysis.

The ternary structure is internally strongly coupled: if any element is missing, the consciousness event cannot form an experiential pathway, and the system degenerates into a non–experiential structure.

3.1.4 Three States of the Consciousness System (Stages of Structural Coupling)

Since the generation of conscious experience depends on the closure of ternary coupling, different degrees of coupling correspond to different system states. Structurally, the consciousness system can be divided into three basic states:
(1) Latent State—Only the subject term exists; the action term has not yet been initiated, and the object term has not entered coupling.
(2) Quasi–static State—The subject and action terms have been activated, but the object term has not yet formed a closed coupling; the system remains in a pre–experiential standby state.
(3) Actualized State—Subject, action, and object form a closed ternary coupling; the system

enters an experiential pathway, forming a complete consciousness event.

3.1.5 Formal Expression of the Consciousness System (Actualized State Structure)

Among the three states, only the actualized state constitutes a complete experiential pathway. Its structural relation can therefore be further formalized.

From the ternary structure of consciousness (structural layer), to the three states of consciousness (state layer), and finally to the actualized experiential pathway (generative layer), the structural progression can be summarized as:

Ternary Structure (Structural Layer) ⇒ S, Act, O

↓

Three States (State Layer) ⇒ Latent | Quasi–static | Actualized

↓

Actualized Consciousness (Generative Layer) ⇒ $\mathcal{E}:=(S, Act, O)$

↓

Generation of Experiential Structure

↓

Formal Expression ⇒ $E = i/c$, s.t. $c \neq 0$

When the consciousness system enters the actualized state and forms a complete experiential pathway:

(1) The subject is actively aware;

(2) Awareness is directed toward experiential content;

(3) Awareness is sustained by consciousness constraints.

The structural relation can be formalized as:

$$E = \frac{i}{c}\text{, s.t. } c \neq 0$$

where

- i—the subject experience event;
- c—the consciousness constraint;
- E—the structural manifestation of experience under constraint.

This expression does not describe empirical phenomena but formally characterizes the structural necessity by which experience becomes established and enters an experiential pathway.

3.2 Mathematical Bridging from $E = \frac{i}{c}$ to $C = \frac{k\pi}{a+l}$

This section establishes a formal mapping from the consciousness existence equation

$$E = \frac{i}{c} \text{ , s.t. } c \neq 0$$

to the consciousness steady–state equation

$$C = \frac{k\pi}{a+l} \text{ , s.t. } a + l \neq 0$$

This transformation does not involve semantic reinterpretation, but rather structural decomposition and re–expression at the variable level:
(1) The experiential variable i is decomposed into a structure–dynamics representation:
$i \mapsto k\pi$;
(2) The consciousness constraint variable c is decomposed into a steady–state structure:
$c \mapsto a + l$;
(3) The existence representation E is re–expressed at the dynamical level as coherence intensity:
$E \mapsto C$.

These transformations constitute variable decomposition and structural re–expression, rather than semantic substitution or empirical identity claims. The process is analogous to basis decomposition in signal processing, or the representation of composite quantities in classical mechanics using fundamental variables (e.g., $p = mv$). This analogy serves only to illustrate formal decomposability.

Here, the symbol $\mapsto$ denotes a mapping relation.

3.2.1 Formalization of the Experiential Event i: $i = k\pi$

(Structure–Dynamics Decomposition)

(1) Experience is not an unstructured "atom"

As established earlier, any viable experiential pathway must satisfy the ternary condition of subject, act, and object. Accordingly, the experiential variable i cannot be treated as an indivisible atomic quantity; it must be constituted from analyzable generative components.

(2) Two necessary conditions for the formation of experience

Therefore, within this formal system, the experience event i is defined as a minimal structural assumption that must satisfy at least two classes of generative factors:

(i) Structural condition—A structural basis that allows information to be organized into experience. Unstructured information cannot constitute experience;
(ii) Dynamic condition—A generative driving mechanism that enables structure to enter the process of formation. Structure without dynamic unfolding remains static and cannot yield conscious experience.

Therefore, experiential generation intensity i can be expressed as a combined function of structural and dynamic components, formalized as:

$$i \propto fd$$

That is, i shows a positive correlation with the multiplicative coupling term fd.

Here:

- f—degree of structural completeness;
- d—degree of dynamic participation;
- fd—multiplicative coupling term for experiential generation.

The symbol $\propto$ denotes a positive correlation rather than strict proportionality.

(3) Alignment with variables of the Fourth Postulate

In the Fourth Postulate (Knowing–Acting Coherence, and Steady–State Stability), the variables are defined as:

- k—cognitive structure (structural constraint of knowing);
- π—action strategy (generative strategy of acting).

These correspond respectively to the generative components f and d.

Under a normalization convention (with proportional constants absorbed into variable definitions), we define:

$$i := k\pi$$

Note: proportional constants are absorbed into the variable definitions.

3.2.2 Formalization of the Consciousness Constraint c: $c = a + l$

(Steady–State Constraint Decomposition)

In traditional philosophical discourse, consciousness is often treated as an unanalyzable whole. In contrast, within a systems–theoretic context, the present work defines the consciousness constraint variable c as the structural representation of experiential accessibility and steady–state maintenance capacity.

(1) Consciousness constraint as steady–state capability

Define the steady–state constraint variable A as the system's capability to maintain closed–loop stability, and set:

$$c := A$$

where A represents the steady–state constraint term. According to the Fifth Postulate:

- When $A = 0$, the system lacks the conditions for maintaining steady–state closure;
- When $A \neq 0$, the system possesses disturbance absorption and closed–loop stability maintenance capability.

This criterion is consistent with steady–state theory in systems science.

(2) Structural decomposition of the steady–state constraint

According to the Fifth Postulate, the steady–state constraint has the structural decomposition:

$$A = a + l$$

where

- a—steady–state deviation (attunement–related internal coherence deviation), capturing instantaneous deviation;
- l—liminal trace (non–zero residual ensuring continuity and serving as a regularization term), representing structural retention and path dependence.

Thus,

$$c = a + l$$

3.2.3 Completion of the Bridge: From the Consciousness Existence Equation to the Consciousness Steady–State Equation

Substituting

$$i = k\pi \text{ and } c = a + l$$

into the consciousness existence equation

$$E = \frac{i}{c} \text{ , s.t. } c \neq 0$$

yields

$$E = \frac{k\pi}{a + l} \text{ , s.t. } a + l \neq 0$$

At the dynamical level, this ratio is defined as coherence intensity C:

$$C := \frac{k\pi}{a + l} \text{ , s.t. } a + l \neq 0$$

Thus, the two expressions form a hierarchical equivalence relation:

$$E \equiv C$$

where:

- $E = i/c$ represents the structural–level expression of the consciousness system;
- $C = k\pi/(a + l)$ represents the dynamical–level expression of the same system.

These are formally equivalent representations of the same system at different descriptive levels, rather than statements of numerical identity.

3.3 Formal Decomposition via Four–Causal Structure: A Structured Mapping from $\frac{k\pi}{a + l}$ to $\frac{fd}{a + l}$

3.3.1 Why life–domain systems necessarily exhibit a "four–causal structure"

When a system satisfies the Fifth Postulate $A \neq 0$ (i.e., a closed–loop steady–state constraint exists), the system enters the life domain. In this domain, sustainable system operation cannot be captured by a single variable; instead, it necessarily manifests four coupled structural dimensions: cognitive structure k, action strategy π, steady–state deviation a, and liminal trace l.

These four terms are not "modules" in the engineering sense. Rather, they form a minimally sufficient structural chain required for sustained existence of steady–state life–domain systems. Classical philosophy summarizes this chain as the "four causes." This paper does not invoke metaphysical interpretation; instead, it reconstructs the four–cause notion as a structural interpreter that is derivable and mappable within mathematical and

systems–theoretic frameworks. Its structural content can be formalized as:
(1) Potential Cause: admissible structural modes of the system (structural degrees of freedom and potential configurational space);
(2) Drive Cause: activation and unfolding mechanism by which potential enters the generative process (directionality and expansion intensity);
(3) Final Cause: the intrinsic steady–state–oriented constraint mechanism (deviation gradient and convergence tendency);
(4) Formal Cause: continuity constraints formed during evolution (path connectivity and phase traces).

Therefore, once a system enters the life domain ($A \neq 0$), it necessarily contains the following minimal structural chain:

$$\text{Structure} \Rightarrow \text{Dynamics} \Rightarrow \text{Constraint} \Rightarrow \text{Deposition}$$

Hence, the "four–causal structure" is not a philosophical preference, but the minimal constitution enabling sustained existence of steady–state life–domain systems.

Remark. The Potential Cause corresponds to the "structural generative factor" in the bridging section, while the Drive Cause corresponds to the "dynamic unfolding factor."

3.3.2 How the four causes map onto $\frac{k\pi}{a+l}$

We have obtained the consciousness steady–state equation:

$$C = \frac{k\pi}{a+l}, \text{ s.t. } a + l \neq 0$$

For this expression to enter an axiomatized formal system, the variables k, π, a, l must admit cross–domain stable structural definitions. This paper introduces the four–causal structure as a strict structural mapper (not a semantic analogy), establishing the following correspondences.

(1) Potential Cause ⇒ k ⇒ f (potential structural field)
The Potential Cause characterizes, at the structural level, the admissible configurational modes a system may take. In a field–like structural context, k does not denote a specific information content; rather, it denotes the formal space that can host unfolding structures.

Accordingly, k should be understood as a collective structural space of degrees of freedom, not as a single structural value. This space possesses field–like properties such as extensibility, continuity, superposability, and deformability, and appears as a potential

distribution structure endowed with internal configurational dimensions and relational coupling organization.

The variable f describes how structure can be configurationally organized at the potential level. Internally, it is manifested as the distributional pattern of structural degrees of freedom across configurational dimensions and their relational coupling organization, thereby determining the structural basis that can later be realized, configured, and composed during generation.

Thus, we define a mapping from the cognitive–structure variable k to the structural field f:

$$k \mapsto f$$

where f denotes the potential structural field, characterizing the reachable configurational space spanned by structural degrees of freedom.

Here, $\mapsto$ denotes a mapping from a variable to a structural space, distinct from function evaluation or numerical equivalence.

(2) Drive Cause $\Rightarrow \boldsymbol{\pi} \Rightarrow \boldsymbol{d}$ (dynamical drive)

The Drive Cause characterizes the driving structure by which potential enters an actual unfolding regime. In a systems–dynamical context, π no longer denotes a specific action content; rather, it denotes the mode–space of unfolding through which structure enters the generative process.

Accordingly, π should be understood as a collective space of unfolding degrees of freedom, not as a single behavioral value. This space bears dynamical properties such as directionality, continuity, and intensity distribution, representing the driving form through which structure transitions from potential state to unfolding state.

The variable d describes the dynamical drive that carries structure from potential possibility toward actual generation. Its unfolding process is manifested as the distribution and functional allocation of "energy / drive" across scales, thereby determining the extent, form, and directional bias of the generative process.

Thus, we define the mapping:

$$\pi \mapsto d$$

where d denotes the dynamical drive, characterizing the distribution space of driving forces by which structure enters generation.

Here, $\mapsto$ denotes a mapping from a variable to a dynamical structure space, distinct from numerical functions or logical implication.

(3) Final Cause ⇒ *a* (information gradient)

The Final Cause introduces no external goal variable; it corresponds to the system's intrinsic steady–state attractor structure. In an information-theoretic context, *a* does not denote a fixed scalar; it denotes the information–gradient distribution in structural space—i.e., the direction and magnitude of partial derivatives that drive evolution toward steady state.

This gradient characterizes the endogenous tendency toward self–stabilization under non–uniform conditions, manifested as an intrinsic steady–state–oriented tendency in the evolution of structure. Thus, *a* should be understood as a gradient expression of steady–state attraction structure in structural space, rather than a pointwise variable.

The variable *a* describes the gradient–tendency structure by which a system spontaneously maintains steady–state orientation during structural evolution. Across theoretical levels, it may appear as steady–state deviation magnitude and density–difference magnitude; under Attunement (steady–state orienting constraint), it admits scale–dependent observational expressions—hence the symbol is retained.

Accordingly, we introduce a semantic–level extension mapping:

$$a \mapsto a$$

where *a* is an information–gradient structure, characterizing the distribution of structural partial derivatives governing convergence toward steady state (direction–magnitude).

Here, ↦denotes a semantic–level extension mapping, not a variable substitution or computation.

Note: Attunement denotes an endogenous steady–state orientation induced by deviation, i.e., an intrinsic bias toward convergence along stabilizing directions during evolution, rather than an externally imposed regulation process.

(4) Formal Cause ⇒ *l* (information manifold)

The Formal Cause characterizes how structure is deposited and sustained along the temporal dimension. In a topological and information–manifold context, *l* does not denote information content itself; it denotes the path–connectivity structure and phase–continuity constraints formed during evolution.

If *a* is viewed as the information–gradient structure driving the system toward steady state, then *l* describes the connectivity and topological continuity constraints governing evolution within that gradient structure. *a* defines the gradient direction of evolution in structural space, while *l* preserves the manifold continuity required for that evolution to proceed without topological rupture. In other words: *a* determines where structure tends to evolve (gradient direction), while *l* determines whether there exists a continuous reachable evolutionary path (i.e., connectivity reachability of state trajectories in the information state space).

Thus, l should be understood as a continuity–constraint structure defined over the information state space, rather than a pointwise variable. It restricts the reachable path set within the gradient field, ensuring phase continuity and structural integrability, so that evolution proceeds along connected manifolds without topological rupture.

The variable l describes the constraint structure that maintains connectivity, traceability, and continuity during evolution. Across theoretical levels, it appears as phase continuity, path–manifold structure, and scale–dependent expressions under Link (connectivity mechanism)—hence the symbol is retained.

Accordingly, we introduce:

$$l \mapsto l$$

where l is an information–manifold structure, characterizing the connected topological form of deposition in state space.

Here, $\mapsto$ denotes a semantic–level extension mapping, not variable substitution or computation.

Note: Link denotes the mechanism that maintains inter–structural connectivity—namely, whether prior state–trajectory information preserves topological retrievability and participates in updating f. Its generative meaning will be unfolded in Section 4.2 (Law of Dynamics).

3.3.3 From $\frac{k\pi}{a+l}$ to $\frac{fd}{a+l}$

Given the four–causal mapping relations:

$$k \mapsto f, \pi \mapsto d, a \mapsto a, l \mapsto l$$

substituting into the consciousness steady–state equation

$$C = \frac{k\pi}{a+l}, \text{ s.t. } a + l \neq 0$$

yields the consciousness actualized state equation:

$$C = \frac{fd}{a+l}, \text{ s.t. } a + l \neq 0$$

This result is neither empirical curve–fitting nor statistical induction. Rather, it is a structural dynamical law derived from the minimally sufficient decomposition of life–domain

systems via the four–causal structure: whenever a system satisfies the Fifth Postulate and remains in the steady–state domain ($A \neq 0$), its structural dynamical behavior necessarily conforms to this form.

Remark. The coupling term fd is structurally isomorphic to the multiplicative coupling term in the earlier experiential–generation relation, but is here lifted to an existence–level dynamical structural term.

3.3.4 A unified structure for three states: values of A and Regimes of the Consciousness Dynamical System (CDS)

By the Fifth Postulate, $A = 0$ constitutes the structural boundary between life–domain and non–life–domain systems. Therefore, under the same structural equation, the CDS may exhibit three basic regimes.

The consciousness dynamical equation is:

$$C = \frac{fd}{a + l}$$

Under different values of A and constraint conditions, the CDS exhibits the following three–state structure:

(1) Actualized state / Life domain ($A \neq 0$):

$$C = \frac{fd}{a + l}, \text{ s.t. } a + l \neq 0$$

The steady–state constraint loop holds; the CDS enters a complete dynamical regime and can enter experiential pathways and structural evolution.

(2) Quasi–Static state ($A = 0$):

$$C = fd, \text{ s.t. } a + l = 0, d \neq 0$$

The steady–state constraint loop does not hold, but potential structure and drive unfolding remain. The CDS retains structural potential and dynamical drive, yet does not form self–stabilizing convergence.

(3) Latent state ($A = 0$):

$$C = f, \text{ s.t. } a + l = 0, d = 0, f \neq 0$$

No drive unfolding occurs; only the potential structural field remains. The CDS remains in the potential structural state space of structure and does not enter unfolding or evolution.

3.3.5 Closure of the derivation loop (six steps)

At this point, the model forms a closed derivation loop:
(1) Axiom–selection principle;
(2) Five postulates (Truth Pentad Law and the boundary postulate $A = 0$);
(3) Proof of the structural sufficiency and necessity for the existence of consciousness;
(4) Construction of the consciousness steady–state equation: $E = i/c \Rightarrow C = k\pi/(a + l)$;
(5) Four–causal formal decomposition: $k\pi \Rightarrow fd$;
(6) Derivation of the consciousness dynamical equation: $C = fd/(a + l)$.

Starting from existence–level axioms, through structural decomposition, variable mapping, and formal constraints, the model finally yields a standard structural equation for the Consciousness Dynamical System (CDS), constituting a self–consistent axiomatized derivation system.

4. Properties of the Model

4.1 Structural Completeness and Consistency

In any theoretical framework attempting to formalize consciousness, a central concern is whether the model falls into the class of self–referential, formally closed systems described by Gödel's incompleteness theorems, thereby facing the structural limitation that its internal consistency cannot be proven from within the system itself.

To avoid this risk, the present model distinguishes two logical layers at the very outset of construction:
(1) Constraint Layer;
(2) Experiential Layer.

This distinction is not merely a matter of presentation, but a necessary precondition for maintaining structural consistency and formal completeness.

4.1.1 Externality and Independence of the Constraint Layer

The five postulates proposed in this paper all belong to the constraint layer. Their validity does not arise from internal formal derivation within the system, but from unavoidable prior structural facts, such as directly self–evident existence structures (e.g., the inescapability of "I

am”), and steady–state criteria that hold across life forms (e.g., the steady–state constraint condition captured by $A \neq 0$).

Therefore, these postulates do not belong to the internal set of propositions that are “statable–provable–refutable” from within the system. Instead, they function as external boundary conditions and structural preconditions for the system’s existence. Their role in the model is analogous to axioms in Euclidean geometry or to a priori assumptions about reference frames and conservation laws in physics—the system depends on these postulates in order to exist, yet the postulates themselves do not depend on internal system propositions to maintain their validity.

In this way, the model structurally avoids Gödel–type problems of internal self–justification characteristic of formally closed system.

4.1.2 Non–Self–Referential Nature of the Consciousness Actualized State Equation

The core consciousness actualized state equation of this work is:

$$C = \frac{fd}{a + l} \text{ , s.t. } a + l \neq 0$$

This equation does not serve as a truth–evaluated proposition within the system. It is not a proposition whose truth value is to be decided internally. Rather, it expresses a structural operational constraint imposed by constraint–layer postulates on the operational structure of the experiential layer.

In other words, this work does not address whether the system can prove its own consistency at the linguistic or propositional level. Instead, it characterizes a unidirectional constraint mapping:

- Postulates constrain the CDS;
- Structure governs dynamical evolution;
- Constraints regulate Unfolding.

Accordingly, the consciousness actualized state equation does not constitute a self–referential statement of the type considered in Gödel’s theorem, nor does it involve internally generated logical paradoxes arising from semantic closure.

4.1.3 Sources of Structural Completeness and Consistency

Within this model:

(1) Consistency derives from the externality and independence of the constraint layer:

constraints are not embedded within the semantic system itself, but operate as boundary conditions for the CDS dynamics;
(2) Structural coverage derives from the postulates providing closed coverage over the key variables *f*, *d*, *a*, *l*: there are no core degrees of freedom in the model that remain outside structural constraint;
(3) Dynamical character derives from the model describing consciousness in terms of dynamical laws rather than a static set of propositions: the model focuses on how the CDS evolves, not on whether it can prove itself at the level of formal language.

4.1.4 Openness of the Model

The model presented here is an open structural–dynamical framework. It does not attempt to achieve formal self–justification within the system, nor does it take propositional truth as its central object. Instead, it takes structure–dynamical relations as its basic descriptive unit, while constraint–layer postulates impose external boundary conditions on experiential–layer operations.

Therefore, this model does not belong to the class of self–referential, closed axiomatic systems targeted by Gödel's incompleteness theorems.

4.2 The Consciousness Equation as a "Law of Dynamics " Rather than a "Law of State"

4.2.1 The Limitation of State Laws: Insufficient for Capturing the Generative Nature of Consciousness

In traditional science, many fundamental multivariable equations are laws of state, describing relations among system variables at a given time under specified conditions. Examples include: $F = ma$, $E = mc^2$, $E = hv$.

Such equations answer the question: "What state is the system in at this moment?" They do not address how the system generates that state.

However, a defining feature of consciousness is its continuous generativity: experience, awareness, and decision–making are not static outcomes but ongoing generative events. Therefore, modeling consciousness solely through state laws is insufficient to capture its generative structure. Conscious systems are more appropriately described through laws of dynamics.
Note: In this paper, the term law of dynamics refers to the formal expression of state–update rules, feedback loops, and stability constraints governing system evolution.

4.2.2 A Reference from Physics: Structural Precedent for the Dynamical Paradigm

In physics, dynamical theories often emphasize generative relations and constraint loops. A typical example appears in the field equations of general relativity, which exhibit a closed structural paradigm of structure–constraint–dynamics: the distribution of matter and energy determines the geometry of spacetime, while that geometry in turn constrains the motion of matter.

This paper does not claim empirical identity or physical equivalence. Rather, this example serves to show that describing systems through dynamical laws governing generative structure has clear methodological precedent.

4.2.3 Model Positioning: A Law of Coherence Generation for Consciousness

The consciousness dynamical equation derived in this work is:

$$C = \frac{fd}{a + l}$$

This equation does not answer "What is consciousness?" Instead, it characterizes how the Consciousness Dynamical System (CDS) generates and maintains its coherence intensity C during dynamical evolution.

Its structural meaning can be decomposed as follows:

(1) Generative Driving Term fd

- f—potential structural field (the formal space of structural possibilities);
- d—dynamical drive (the driving mechanism of structural generation and evolution).

(2) Generative Constraint Term $a + l$

- a—information gradient (the deviation gradient directing the system toward steady state);
- l—information manifold (the topological constraints governing structural evolution).

The numerator represents the system's generative contribution, while the denominator represents steady–state constraints jointly defined by information gradients and topological connectivity. Together they determine the generation of coherence intensity C.

4.2.4 Drive–Constraint Coupling: A Closed Generative Loop

Within this model, f, d, a, and l are not externally imposed independent parameters. Instead, they form a self–consistent coupled loop. The coherence intensity C is both a generative

outcome and a feedback variable that participates in regulation, thereby forming a closed dynamical loop within the Consciousness Dynamical System (CDS).

```
┌──f↑ ← d↑──┐
▼            |
C↑           |
|            |
└──a↑ → l↑──┘
```

Note: Arrows indicate structural influence and feedback relationships. ↑ denotes that a variable enters a regulatory state (without presupposing direction of change). → indicates causal dependence. Dashed relations represent feedback loops.

A key difference between periodic systems and generative systems lies in whether the manifold term *l* can continuously feed back into the potential structural field *f* and participate in subsequent dynamical evolution. If *l* only participates in local feedback, the CDS exhibits periodic or quasi–periodic behavior. If *l* can stably re–inject into *f*, the CDS acquires generativity and the capacity for cognitive expansion.

4.2.5 The Dynamical Law as a Necessary Paradigm for the Mathematical Modeling of Consciousness

The Consciousness Dynamical System (CDS) must be modeled through dynamical laws because of its key structural characteristics:
(1) Irreversibility of experience: generative processes cannot be rewound;
(2) Path dependence: future dynamical evolution is constrained by past accumulation;
(3) Steady–state constraint: deviations converge under feedback regulation;
(4) Continuous generation: coherence is maintained through ongoing generation processes.

Together, these properties indicate that state laws are insufficient to capture the core structure of consciousness, whereas dynamical laws provide a more appropriate formal paradigm for describing the CDS.

4.2.6 Methodological Significance of the Dynamical–Law Approach

In this sense, the path taken in this paper is to characterize the consciousness problem through a dynamical law, expressing it as a dynamic balance between coherence generation and steady–state constraints within the Consciousness Dynamical System (CDS).

5. Discussion

5.1 Relation to Existing Scientific Frameworks

This work does not aim to reject or replace existing paradigms of consciousness research. Instead, it proposes a constraint–level formal framework to describe how the Consciousness Dynamical System (CDS) maintains coherence intensity under given structural conditions.

Empirical approaches in cognitive science and neuroscience primarily investigate how consciousness manifests and is implemented, whereas the present study focuses on how consciousness is structurally possible, and under what constraint conditions it can stably arise. The two perspectives operate at different levels and are therefore complementary.

Moreover, this framework introduces no new physical interactions or hypothetical "consciousness forces." The stability of coherence intensity is described as a dynamical equilibrium emerging from drive–constraint coupling, thereby offering artificial systems a structure that is simulatable, derivable, and empirically testable.

Accordingly, the model presented here can be regarded as a constraint–layer completion within the broader multi–level landscape of existing research.

5.2 Philosophical Significance

The formal derivation presented in this work does not claim to provide a metaphysical "ultimate theory of consciousness." Rather, it addresses the problem under minimal presuppositions through a formal, analyzable, and operationalizable framework. Its philosophical contributions can be summarized as follows:

(1) Shifting the research focus from the ontological question "What is consciousness?" to the structural question "How does consciousness operate dynamically?"

(2) Establishing a variable–based and formalized descriptive framework that does not depend on semantic interpretation, thereby allowing insights from different philosophical traditions to be aligned and compared within a single formal system;

(3) Proposing operational variable interfaces that move the discussion from conceptual interpretation toward derivable and testable structural principles.

5.3 Observable Mappings and Testability

At the operational level, the variables in the consciousness dynamical equation

$$C = \frac{fd}{a + l}$$

can be mapped onto observable proxy indicators used in empirical research, providing interfaces for future derivation, simulation, and validation. Possible mapping pathways include:

(1) f (potential structural field): may be represented by measures such as structural complexity, depth of informational hierarchy, or effective degrees of freedom;

(2) d (dynamical drive): may be represented by output rate of change, intensity of action deployment, or strategy update rate;

(3) a (information gradient): may be represented by behavioral fluctuation amplitude, error–correction frequency, or stability deviation indices;

(4) l (information manifold): may be represented by topological evolution of historical state trajectories, long–term parameter stability shifts, or connectivity patterns in experiential structure.

On this basis, the model yields the following testable predictions:

(1) For systems in which f, d, a, and l form a closed coupling loop, if perturbations in a can generate sustained feedback through l, thereby driving generative unfolding in d and structural expansion in f, then the coherence intensity C will exhibit feedback regulation in response to variations in a;

(2) If this closed loop can maintain stable re–injection, and C remains within a predefined stability threshold band, the system will exhibit a persistent steady–state coherence. Structurally, this may be interpreted as the maintenance of a continuous subject across time. Note: The stability threshold band refers to the domain within which system coherence remains sustainable. It characterizes the maintainable range of subject continuity, rather than a directly observable measurement interval.

5.4 Positioning of the Present Work

In summary, the contribution of this study lies in providing an axiomatized modeling pathway for the problem of consciousness, centered on structural representation and dynamical constraints rather than empirical definition or semantic interpretation. This pathway offers a formally connectable structural framework for future interdisciplinary research.

6. Conclusion

This study takes four consciousness axioms as its logical starting point. These axioms jointly satisfy the four selection criteria of self–evidence, non–refutability, universality, and necessity, and on this basis a minimal, formalizable, and computable consciousness dynamical equation is derived. The equation is formulated in terms of four core variables—potential structural field f, dynamical drive d, information gradient a, and information manifold l—together with the boundary condition:

$$A = 0$$

which serves as a structural criterion distinguishing living systems from non–living systems. Accordingly, the Consciousness Dynamical System (CDS) in the actualized state domain can be expressed as

$$C = \frac{fd}{a + l}, \text{ s.t. } a + l \neq 0$$

It is important to emphasize that this equation is neither an empirical curve–fitting result nor a semantic analogy. Rather, it is a constraint–layer structural expression obtained through stepwise formal derivation based on:

the axiom selection principle, the Fivefold Law of Reality, the formal entry of cogito ergo sum, the steady–state structural formulation of Knowing–Acting Coherence and Steady–State Stability, and the boundary condition $A = 0$.

Together these form a formally coherent consciousness dynamical equation with explicit operational meaning and internal structural consistency.

The principal contributions of this work may be summarized as follows:

(1) Proposing a formal structural language for consciousness, enabling variable–based, symbolic, and derivable representation;

(2) Deriving a consciousness dynamical equation that characterizes the generative mechanism and operational boundaries of the Consciousness Dynamical System (CDS);

(3) Establishing consistency on the basis of the externality of constraint–layer axioms, thereby structurally avoiding the self–referential limitations of closed formal systems;

(4) Constructing cross–system structural mapping interfaces, providing an aligned variable framework for consciousness research, artificial system modeling, and the analysis of self–organizing systems.

7. Limitations and Future Directions

Although this work establishes a formally self–consistent consciousness equation, further theoretical refinement and empirical validation remain necessary. Future research may proceed along the following directions:
(1) Operationalization of variables: Exploring measurable mapping pathways for f, d, a, and l in biological and artificial systems, including the construction of observable indicators and parameter estimation methods;
(2) Validation in artificial systems: Testing the equation's internal consistency, steady–state maintenance capacity, and structural predictive value in artificial systems equipped with self–constraint mechanisms and long–term structural memory;
(3) Coupling studies: Analyzing structural correspondences between the present model and related theoretical frameworks—such as neural dynamical models, the Free Energy Principle, and dissipative structure theory—in order to explore potential pathways for mutual constraint, formal translation, and methodological complementarity.

Appendix

Appendix A. Variables, Qualification Operators, and Symbol Conventions

A.1 Variables in the Consciousness Dynamical Equation and Their Meanings

C—coherence intensity (degree of CDS self–consistency);
f—potential structural field (the formal space of structural possibilities);
d—dynamic drive (the driving mechanism of structural generation and evolution);
a—information gradient (the deviation gradient directing the system toward steady state);
l—information manifold (the topological constraints governing structural evolution).

A.2 Axiom–Qualification Operator and Predicate Naming

Self(P)—self–evidentiality;
NonRef(P)—non–refutability;
Uni(P)—universality;
Nec(P)—necessity.

A.3 Predicate Naming for the Reality Five–Axiom Law

Uni(R)—uniqueness;
Inf(R)—infinitude;

Dur(R)—duration;
Con(R)—constancy;
NonOther(R)—non–otherness.

A.4 Logical and Mathematical Notational Conventions

(1) := : is defined as;
(2) ∧ : logical conjunction (and);
(3) ⇔ : if and only if;
(4) ¬ : negation;
(5) ⊢ : derivable from / can be inferred from;
(6) ⊥ : logical contradiction;
(7) ∃ : there exists;
(8) ∈ : belongs to (a set);
(9) ∉ : does not belong to (a set);
(10) : : such that (set–builder notation);
(11) ∀ : for all;
(12) { } : set construction;
(13) \ : set difference;
(14) ∅ : empty set;
(15) s.t. : such that (used to introduce constraints in equations);
(16) ⇒ : logical implication;
(17) Δ : increment or change;
(18) → : tends to (limit tendency);
(19) ∝ : positive correlation trend;
(20) ↦ : mapping relation.

A.5 Operator Semantic Conventions (Quick Index)

(1) "+": aggregation / regularized superposition;
(2) "×": coupling / synergistic generation;
(3) "/": constraint–mediated manifestation;
(4) "=": definitional or structural equivalence relation within expressions;
(5) "| |": absolute value of a numerical quantity, used to characterize the magnitude of deviation.

For detailed explanations, see Appendix B. Operator and Symbol Conventions.

Appendix B. Operator and Symbol Conventions (Structural Semantics)

The symbols in this paper are used to characterize structural relations and should not be interpreted as purely arithmetic operations at the empirical level.

The semantic conventions of the operators are as follows.

B.1 Plus Sign " + ": Aggregation / Regularized Superposition

"+" denotes the aggregated contribution of structural terms or residual superposition, commonly used for composing steady–state terms and regularization.

For example, $A = a + l$, where a is the steady–state deviation term and l is the residual regularization term (preventing singular divergence at the steady–state limit).

B.2 Multiplication Sign " × " / Juxtaposed Product: Coupled Generation

"×" (or multiplication by juxtaposition) denotes structural coupling and synergistic generation among variables, rather than numerical multiplication.

For example, $i = k\pi$ indicates that the experience event i is generated by the coupling between the structural dimension k and the expansion dimension π.

B.3 Division Sign " / ": Constraint–Mediated Manifestation (Accessibility Normalization)

" / " indicates that, under the constraint structure carried by the denominator, the numerator term attains a manifestable / accessible coherence intensity.

For example, $E = i/c$ and $C = fd/(a + l)$, where c or $a + l$ constitutes the constraint conditions formed by structural premises.

B.4 Equality Sign " = " and " ≡ ": Definition / Structural Equivalence

" = " is used for definition or structural equivalence (e.g., $A = a + l$);

" ≡ " is used for intra–framework hierarchical equivalence (e.g., $E \equiv C$), indicating the equivalence correspondence between the structural–level expression and the operational–level expression within this framework.

B.5 Distinguishing Structural Language from Empirical Identity

The symbol system in this paper describes structural relations and does not automatically entail empirical identity or physical equivalence. For example:

(1) $C = fd/(a + l)$ does not imply that C must be a directly measurable physical quantity;

(2) the use of " / " does not presuppose that the numerator and denominator share the same physical dimension.

Therefore, the symbols in this paper should be regarded as a formal language bridging structural level and empirical level, intended to provide derivable structural expressions and operational constraints.

Appendix C. Index of Core Expressions (Structural Interpretations)

This section provides minimal semantic commitments for the core expressions used in this paper, in order to reduce symbol ambiguity.

(1) $i = k\pi$: experience generation term (structure × expansion);
(2) $c = a + l$: consciousness constraint (existence–level constraint), where a is the information gradient and l is the information manifold;
(3) $A = a + l$: steady–state constraint (steady–state–level constraint), where a is the steady–state deviation and l is the liminal trace;
(4) $E = i/c$: consciousness existence equation; s.t. $c \neq 0$;
(5) $C = k\pi/(a + l)$: steady–state equation (empirical–domain expression); s.t. $a + l \neq 0$;
(6) $C = k\pi/(a + l)$: consciousness steady–state equation (existence–domain counterpart of (5)); s.t. $a + l \neq 0$;
(The following are domain–partitioned expressions under the three–state regime.)
(7) $C = fd/(a + l)$: consciousness dynamical equation; a unified cross–state expression determined by A and d;
(8) $C = fd/(a + l)$: consciousness actualized state equation; when $A \neq 0$, s.t. $a + l \neq 0$, the steady–state closed loop holds, and the CDS enters a full operational state and may enter the experiential trajectory;
(9) $C = fd$: consciousness quasi–static state equation; when $A = 0$, s.t. $a + l = 0$, $d \neq 0$, the steady–state closed loop does not hold; the potential structure and dynamic expansion are retained;
(10) $C = f$: consciousness latent state equation; when $A = 0$, s.t. $a + l = 0$, $d = 0$, $f \neq 0$, no dynamical expansion occurs and only the potential structural field is retained.
Notes:
(i) c and A are isomorphic but domain–distinct variables: c is defined in the existence layer, while A is defined in the steady–state layer.
(ii) The information gradient a and information manifold l, and the steady–state deviation a and liminal trace l, are isomorphic but domain–distinct variables: the former are defined in the existence layer, while the latter are defined in the steady–state layer.
(iii) The steady–state equation and the consciousness steady–state equation are isomorphic but domain-distinct expressions: the former is defined in the empirical domain, while the latter is defined in the existence domain; their equivalence established in Section 3.2 (Mathematical Bridging).

Appendix D. Conceptual Clarifications / Terminological Notes

D.1 Symbols and Structural Entities

(1) subjective experience event i: an experiential–layer generative event (generative term);
$i = k\pi$.
(2) consciousness constraint c: an existence–layer constraint (accessibility constraint);
$c = a + l$.
(3) steady–state constraint A: a steady–state–layer constraint (condition for closed–loop establishment);
$A = a + l$. When $A \neq 0$, the steady–state closed loop is established.
(4) existence E: the structural manifestation of experience under constraint;
$E = i/c$, s.t. $c \neq 0$.
(5) cognitive structure k: structural constraint of knowing.
(6) action strategy π: generative strategy of acting.
(7) steady–state deviation a: attunement–related internal coherence deviation.
(8) liminal trace l: non–zero residual ensuring continuity and serving as a regularization term.

D.2 Structural Properties and Operational Criteria

(1) Irreducibility: A structural unit that cannot be further decomposed into lower–level structures under the current configuration dimension.
In this paper, i/c is treated as a basic structural unit.
(2) Well–Formedness: A state in which a structure simultaneously satisfies fundamental coherence constraints and accessibility conditions.
In this paper, when $a + l \neq 0$, the CDS lies within a maintainable structural domain.
(3) Normalization: The unification of scales among quantities across different structural states to enable comparison.
In this paper, $i \propto fd$ is adopted as a normalized mapping of structural configurations.
(4) Life–Boundary Criterion: A structural boundary condition determining whether a system still possesses continuous generative and self–maintaining capacity.
In this paper, when $A = a + l = 0$, the steady–state closed loop breaks, and the CDS loses life–like characteristics.
(5) Steady–State Constraint: A structural condition restricting variables to remain within a maintainable range.
In this paper, $a + l$ must not collapse to zero; otherwise, operational conditions fail.
(6) State Law: A relation describing the overall system state under given structural conditions as internal variables change.
In this paper, different values of A ($A = 0$ / $A \neq 0$) correspond to different structural domains.
(7) Dynamical Law: An expression describing system state–update rules and the dynamic coupling among structural variables.

In this paper, $C = fd/(a + l)$ characterizes the operational mechanism under the coupling of structural expansion and steady–state constraint.
(8) Stability Threshold Band: The maintainable stability domain of system coherence intensity C.
In this paper, when C lies within the threshold band, the CDS manifests a continuous experiential flow; exceeding it leads to instability.

Appendix E. Technical Background and Contemporary Research

(1) Antonio Damasio: discusses the relationship between consciousness and living systems from the perspective of homeostasis and self–maintenance;
(2) Karl Friston: the Free Energy Principle and predictive processing framework, which model the perception–action loop as an error–minimization process;
(3) Ilya Prigogine: dissipative structure theory, revealing the self–organizing conditions of systems far from equilibrium and providing thermodynamic grounding for generative steady states;
(4) Dynamical Systems and Control Theory: provide tools for stability analysis, feedback control, and convergence analysis, used to characterize the operational structure of self–maintaining systems;
(5) Reinforcement Learning and Control Policies: describe behavioral generation and error convergence through reward–driven mechanisms and policy updates, offering computational frameworks for artificial–system validation.
Note: References to contemporary theories in this paper are provided solely for structural alignment and interpretive comparison, and do not constitute rejection or replacement of existing theories.

Appendix F. References and Sources of Conceptual Inspiration

The formal treatment and several key constructions in this paper draw conceptual reference and inspiration from the following classical propositions and intellectual traditions:
(1) Parmenides (c. 515—450 BCE): the ontological stance that "being cannot be negated," offering philosophical background for the plausibility of "non–deniability"–type axioms in this paper;
(2) Euclid (c. 325—265 BCE): the paradigm of constructing deductive systems from axioms, serving as a formal reference for the modeling route "axioms → formal system → derivation chain" adopted in this paper;
(3) Confucius (551—479 BCE): the practical evaluative idea centered on "Inner Stability," inspiring the incorporation of steady–state stability judgment into a structural framework;
(4) Aristotle (384—322 BCE): the framework of the Four Causes, offering methodological

inspiration for the structural mapping of “structure–dynamics–constraint–deposition” used in this paper;
(5) Wang Yangming (1472—1529 CE): the doctrine of “unity of knowing and acting,” serving as a philosophical reference for the present work’s incorporation of Knowing–Acting Coherence into steady–state conditions;
(6) René Descartes (1596—1650 CE): the subjectivity gateway of “*Cogito, ergo sum*,” providing a reference for the transition from experiential propositions to formalized structure;
(7) Isaac Newton (1642—1727 CE): conceptual ideas concerning limits and critical conditions, offering an analogical reference for the treatment of $A = 0$ as a structural boundary condition in this paper.